

D075

Information Technology Management Essentials

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A clinic group with nine locations that already uses cloud email is replacing its scheduling system. The team is interviewing front-desk staff and documenting what the new system must do. Which SDLC phase is this?

- A. Maintenance
- B. Implementation
- C. Design
- D. Analysis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which activity belongs in the design phase of the SDLC?

- A. Interviewing users about their needs
- B. Installing the system and training users
- C. Fixing bugs reported after launch
- D. Creating the database schema and screen layouts

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An Orders table with 2.3 million rows has the fields OrderID, OrderDate and CustomerID. CustomerID matches the ID field in the Customers table. The database server was upgraded last year. What is CustomerID in the Orders table?

- A. The primary key
- B. A unique field, since every ID is always unique
- C. A foreign key
- D. A composite key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which field is the best primary key for an Employees table?

- A. HireDate
- B. LastName
- C. EmployeeID
- D. DepartmentID

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A small distributor's Orders table repeats each customer's full address on every order row. The sales team prints invoices from it every Friday. When a customer moved, some rows were updated and some were not. What is the best fix?

- A. Move customer data into a Customers table and link orders by CustomerID.
- B. Add a second address column to each order row.
- C. Back up the table more often.
- D. Keep everything in one table; it always performs better.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An HR database has 14 tables, and a new report will run nightly. The report must list departments whose average salary is above \$70,000. Which approach is correct?

- A. GROUP BY department, then HAVING AVG(salary) > 70000
- B. ORDER BY AVG(salary) DESC
- C. WHERE salary > 70000, then GROUP BY department
- D. WHERE AVG(salary) > 70000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In SELECT dept, COUNT(*) FROM Employees WHERE status = 'active' GROUP BY dept, what does the WHERE clause do?

- A. Removes departments after the counts are calculated
- B. Keeps only active employees before they are counted by department
- C. Counts only departments whose name is active
- D. Sorts the departments by status

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A college with 12,000 students on three campuses is designing a database. Each student takes many courses, and each course has many students. How should the relationship be modeled?

- A. Merge students and courses into one table.
- B. Add an Enrollments table holding StudentID and CourseID as foreign keys.
- C. Add a CourseID field to the Students table.
- D. Store a comma-separated list of course IDs in each student row.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A workbook with six tabs, built by a former intern, holds the sales tax rate in cell B1 and prices in A3:A20. The formula =A3*B1 in B3 gives wrong answers when copied down. What should B3 contain?

- A. =SUM(A3*B1)
- B. =A3*\$B\$1
- C. Retype =A3*B1 by hand in each row
- D. =\$A\$3*B1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

The formula =B\$2*\$A5 in cell C5 is copied to cell D7. What formula appears in D7?

- A. =C\$2*\$A7
- B. =B\$2*\$A5
- C. =C\$4*\$B7
- D. =C\$2*\$A5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An office has a 200 Mbps internet plan that costs \$65 a month from a provider rated second in the region. About how long does it take to download a 1 GB file (1,000 MB) at full speed?

- A. About 5 seconds
- B. About 40 minutes
- C. About 0.6 seconds
- D. About 40 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A hospital with a single central billing office is replacing its 15-year-old billing system with one from a vendor based in Minneapolis. Billing cannot be interrupted, and the budget allows extra staff effort during the switch. Which conversion strategy fits best?

- A.** Phased conversion
- B.** Pilot conversion
- C.** Parallel conversion
- D.** Direct cutover

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A clinic group ~~with nine locations that already uses cloud email~~ is replacing its scheduling system. The team is interviewing front-desk staff and documenting what the new system must do. Which SDLC phase is this?

Target word: SDLC phase **Decides:** Documenting what the system must do = requirements.

A	Inverted operation	The last phase, after the system is live.
B	Confusion set	Implementation builds, tests and installs.
C	Adjacent category	Design specifies how the system will meet requirements, after analysis.
D	Correct	Analysis gathers and documents requirements.

2. Answer D Concept 1 • transfer

Which activity belongs in the design phase of the SDLC?

Target word: pick the design activity **Decides:** Design specifies how.

A	Inverted operation	That is analysis, which comes before design.
B	Adjacent category	That is implementation.
C	Confusion set	That is maintenance.
D	Correct	Specifying structure and interfaces is design.

3. Answer C Concept 2

An Orders table ~~with 2.3 million rows~~ has the fields OrderID, OrderDate and CustomerID. CustomerID matches the ID field in the Customers table. ~~The database server was upgraded last year.~~ What is CustomerID in the Orders table?

Target word: key type **Decides:** It points to another table's primary key.

A	Confusion set	OrderID identifies each order; many orders share a CustomerID.
B	Absolute language	A customer appears on many orders.
C	Correct	It links each order to a customer's primary key.
D	Adjacent category	A composite key combines fields to identify rows; that is not its role here.

4. Answer C Concept 2 • transfer

Which field is the best primary key for an Employees table?

Target word: choose a primary key **Decides:** Must be unique and stable for every row.

A	Confusion set	Many employees share a hire date.
B	Dropped element	Drops uniqueness; many employees share last names.
C	Correct	A unique, stable identifier.
D	Inverted operation	That links to Departments; it is a foreign key here.

5. Answer A Concept 3

A small distributor's Orders table repeats each customer's full address on every order row. ~~The sales team prints invoices from it every Friday.~~ When a customer moved, some rows were updated and some were not. What is the best fix?

Target word: fix for an update anomaly **Decides:** Redundant data stored on many rows.

A	Correct	Normalization stores each address once.
B	Inverted operation	Adds more redundancy.
C	Adjacent category	Backups protect against loss, not inconsistency.
D	Absolute language	Denormalizing is a trade-off and causes these anomalies.

6. Answer A Concept 4

~~An HR database has 14 tables, and a new report will run nightly.~~ The report must list departments whose average salary is above \$70,000. Which approach is correct?

Target word: filter on an aggregate **Decides:** The condition is on a group average.

A	Correct	HAVING filters groups after aggregation.
B	Adjacent category	Sorts groups but does not filter them.
C	Dropped element	Filters individuals, dropping the average condition.
D	Confusion set	WHERE runs before grouping and cannot use aggregates.

7. Answer B Concept 4 · transfer

In SELECT dept, COUNT(*) FROM Employees WHERE status = 'active' GROUP BY dept, what does the WHERE clause do?

Target word: role of WHERE **Decides:** WHERE acts on rows before grouping.

A	Inverted operation	That would be HAVING.
B	Correct	Rows are filtered first, then grouped.
C	Confusion set	The condition is on each employee's status, not the department.
D	Adjacent category	Sorting is ORDER BY.

8. Answer B Concept 5

~~A college with 12,000 students on three campuses~~ is designing a database. Each student takes many courses, and each course has many students. How should the relationship be modeled?

Target word: model a many-to-many **Decides:** Many on both sides.

A	Adjacent category	Creates heavy redundancy.
B	Correct	A junction table resolves many-to-many.
C	Dropped element	Allows only one course per student.
D	Confusion set	Breaks first normal form and makes querying hard.

9. Answer B Concept 6

A workbook with six tabs, built by a former intern, holds the sales tax rate in cell B1 and prices in A3:A20. The formula =A3*B1 in B3 gives wrong answers when copied down. What should B3 contain?

Target word: correct reference **Decides:** The tax rate cell must stay fixed while the price row changes.

A	Confusion set	Wrapping in a function does not change how references copy.
B	Correct	Locks the tax rate; A3 still shifts row by row.
C	Dropped element	Drops the reason for copying; B1 would still need to be fixed each time.
D	Inverted operation	Locks the price instead of the rate.

10. Answer A Concept 6 · transfer

The formula =B\$2*\$A5 in cell C5 is copied to cell D7. What formula appears in D7?

Target word: result of copying mixed references **Decides:** Row 2 and column A are locked; the other parts shift by +1 column and +2 rows.

A	Correct	B shifts to C, row 2 stays; column A stays, row 5 shifts to 7.
B	Dropped element	Treats the formula as fully absolute.
C	Inverted operation	Moves the locked parts and would keep the unlocked ones.
D	Confusion set	Shifts the column but forgets the row of the second reference.

11. Answer D Concept 7

An office has a 200 Mbps internet plan that costs \$65 a month from a provider rated second in the region. About how long does it take to download a 1 GB file (1,000 MB) at full speed?

Target word: download time **Decides:** $1,000 \text{ MB} \times 8 = 8,000 \text{ megabits}$; $8,000 / 200$.

A	Dropped element	$1,000 / 200$ skips converting bytes to bits.
B	Confusion set	Right number, wrong unit.
C	Inverted operation	$1,000 / (200 \times 8)$ converts in the wrong direction.
D	Correct	$8,000 \text{ Mb} / 200 \text{ Mbps} = 40 \text{ s}$.

12. Answer C Concept 8

A hospital with a single central billing office is replacing its 15-year-old billing system with one from a vendor based in Minneapolis. Billing cannot be interrupted, and the budget allows extra staff effort during the switch. Which conversion strategy fits best?

Target word: conversion strategy **Decides:** Critical system with budget for extra effort.

A	Confusion set	Moves module by module but leaves each cut-over module without a fallback.
B	Adjacent category	Pilots need several sites; there is only one billing office.
C	Correct	Running both systems keeps a fallback; lowest risk.
D	Inverted operation	Highest risk: no fallback if the new system fails.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	SDLC phases	Planning (scope, feasibility), analysis (what the system must do), design (how it will do it), implementation (build, test, install, train), maintenance.	Mixing up analysis (requirements) and design (specifications).	Gathering user needs vs. drawing screens and schemas.
2	Primary and foreign keys	A primary key uniquely identifies each row; a foreign key is a field that matches another table's primary key to link the tables.	Thinking any ID field is a primary key.	An ID from another table stored in this one.
3	Normalization	Store each fact once; split repeating data into related tables to prevent update, insert and delete anomalies.	Adding columns or backups to fix redundancy problems.	The same data repeated on many rows and updates that miss some.
4	WHERE vs. HAVING	WHERE filters rows before grouping; HAVING filters groups after aggregation.	Putting an aggregate like AVG in a WHERE clause.	A condition on an average, count or sum per group.
5	Many-to-many relationships	A many-to-many relationship is resolved with a junction table holding both foreign keys.	Putting a list of IDs in one field or a single foreign key on one side.	Each A has many B and each B has many A.
6	Absolute vs. relative references	A \$ locks the column or row that follows it when a formula is copied; unlocked parts shift.	Locking the wrong reference, or thinking functions change referencing.	A formula copied down that points to one fixed input cell.
7	Bits vs. bytes	1 byte = 8 bits; network speeds are in bits per second, file sizes in bytes.	Dividing a byte size by a bit rate without converting.	Mbps next to MB or GB.
8	System conversion strategies	Direct = switch at once (highest risk); parallel = run both (lowest risk, highest effort); pilot = one site first; phased = one module at a time.	Choosing direct cutover for a system that cannot fail.	How critical the system is and how many sites or modules there are.

Course scope (public description): Information Technology Management Essentials includes topics such as information systems analysis, database resource management, spreadsheet literacy, and computer literacy concepts. This course will help students understand the importance of information technology in an organization and apply databases to solve business problems.